26900071 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEBOT303 - GREEN TECHNOLOGY AND SUSTAINABLE DEVELOPMENT

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. State two principles of green chemistry related to waste reduction. [U] / [CO1]
2. State the role of a penstock in a small hydro plant. [K] / [CO3]
3. Mention any two SDGs related to environmental sustainability. [K] / [CO3]
4. How does the use of conventional energy affect the availability of natural resources? [K] / [CO3]
5. State the meaning of the 3R approach of waste management. [U] / [CO2]
6. Name one type of waste that can be converted into a useful product. [K] / [CO2]
7. What is the primary objective of EIA? [K] / [CO4]
8. Mention two challenges in implementing green technologies. [K] / [CO5]
9. What is LEED? [K] / [CO5]

10.	Define renewable energy gadget with one example.	[K] / [CO1, CO5]
11.	State any two benefits of eco-mark certification on a product.	[U] / [CO5]
12.	How can renewable energy improve the sustainability of an establishment?	[U] / [CO4, CO5]

[2x10 = 20]

Part B

Short Answer Questions

Answer any **8** questions

Each question carries **5** marks

13.	Define the principle of microwave assisted synthesis. What are the major advantages of thei extraction process?	[U] / [CO1]
14.	Discuss the role of green technology in environment protection.	[U] / [CO1]
15.	How technology contributes to economic sustainability.	[K] / [CO3]
16.	Explain how carbon credits can be associated with emission-reduction activities.	[U] / [CO2]
17.	Explain how phytonanotechnology can improve plant-based remediation.	[U] / [CO2]
18.	Discuss the four major phases of LCA and their significance in environmental assessment.	[U] / [CO4]
19.	Explain the concept and importance of water conservation.	[U] / [CO5]
20.	How local institutions can contribute to environmental sustainability through eco-friendly initiatives.	[U] / [CO1]
21.	Describe any two renewable energy gadgets commonly used in households.	[U] / [CO5]
22.	Explain how green technologies can be incorporated into the design and operation of institutional buildings to improve resource efficiency.	[U] / [CO4, CO5]

[5x8 = 40]

Part C

Essay Questions

Answer any **1** questions

Each question carries **10** marks

23.	Describe the role of green technology in achieving sustainable development in India.	[U] / [CO1]
24.	Discuss the objectives, significance and environmental benefits of Eco-mark certification in India.	[U] / [CO5]

[10x1 = 10]